

Course outline Fall 2024

Course Name: Video game architecture

Course Code: CS-4650

INSTRUCTOR

Instructor(s)	Dr. Hossam Mekawy
E-mail	Hossam.mekawy@uofcanada.edu.eg
Office Location	B211
Office Hours	(by appointment – please send an email a day before to check if the time slot is not already reserved) Monday: 2:30 pm to 3:00 pm, Wednesday: 11:30 to 12:30 pm, Thursday: 11:00 am to 12:00 pm

TEACHING ASSISTANT

Instructor(s)	Eng. Omar Eltouny
E-mail	Omar.Aymen@uofcanada.edu.eg
Office Location	B318
Office Hours	Sunday 1:25 to 4 pm at B224, Tuesday 10:00 to 11:00 am at B318, Monday 8:30 to 9:30 am at B318.

Class Time

Lecture	Lab (Optional)
Sunday 08:30 am - 09:45 am	Sunday 13:25 – 16:00 pm
Monday 11:20 am - 12:35 pm	

Class LOCATION

Lecture	Lab (Optional)
B224	B224

COURSE DESCRIPTION

This programming-driven course aims to explore the various systems that comprise a typical video-game project, including event systems, state machines, rendering, scripting and AI programming. Students will implement these components throughout the course with the end goal of building a small game. Three lectures hours per week

Requisites:

Computer Science 4350 - Must be completed prior to taking this course.

Computer Science 4360 - Must be taken either prior to or at the same time as this course.

It is mandatory to acquire an average grade of 70% in CS-4650, CS-4360, and CS-4810 for students enrolled in the major in computer science with specialization in video game programming in order to enroll in the graduation project (CS-4830 Video game programming Project)

RECOMMENDED TEXTBOOKS

- “Game Engine Architecture” 3rd edition, Jason Gregory, ISBN-13: 978-1-1380-3545-4, CRC Press, 2019
- The lectures and exercises materials would provide sufficient information on Unreal Engine for building the targeted game, yet the students are encouraged to utilize any book or online resources on Unreal Engine 5.

LEARNING OBJECTIVES

LO #1 Demonstrate an understanding of core concepts and design patterns related to game loop, game timing, game objects, events, data-driven design, scripting, game AI, navigation, content pipeline, resource management, game debugging and profiling.

LO #2 Build familiarity with these concepts through exercises and assignments.

LO #3 Utilize the understanding of the game design patterns and their implementation in Unreal Engine 5 to build a complete game.

CLASS SCHEDULE AND GRADING SCHEME

<i>Week</i>	<i>Lecture</i>	<i>(Optional) Lab</i>
1	The Game Loop + dealing with time Unreal engine introduction	Exercises on using timing and game loop
2	Game Objects Unreal engine class hierarchy	Exercises on using game objects and their hierarchy
3	Events	Exercises on different design patterns in events

4	Data driven design	Exercises on utilizing data driven
5	Midterm exam	
6	scripting	scripting exercises
7	State Machines	state machines exercises
8	Collision and Physics	Collision exercises
9	Game AI	Game AI exercises
10	Navigation	Navigation exercises
11	Content Pipeline/ Resource management/ debugging and profiling	Content Pipeline exercises
12	Project delivery/presentation	project support
13		

GRADING SCHEME

<i>Assessments</i>	<i>Grade</i>	<i>Date</i>
<i>Lecture participation</i>	10%	All lectures
<i>Assignment 1</i>	10%	Week 3
<i>Quiz</i>	5%	Week 4
<i>Midterm</i>	10%	Week 5
<i>Assignment 2</i>	10%	Week 7
<i>Assignment 3</i>	10%	Week 9
<i>Project</i>	10%	Week 10
<i>Final Exam</i>	35%	

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HOMEWORK/ASSIGNMENTS

- You are expected to complete all weekly reading in preparation of class. Occasionally I may also hand out articles or cases not in the text that are required for you to read and prepare for class discussion. In addition, you will have individual/group assignments and term-based reports and assignments throughout the term. **ALL REQUIRED COURSE WORK MUST BE SUBMITTED TO PASS THE COURSE.**
- All reports required during this course shall be original and following academic integrity policy. The reports will be submitted on Turnitin tool, Turnitin check for similarities among other reports and the internet. If similarity beyond reasonable amount is detected penalties extending to filing an academic dishonesty case may be applied
- All students are expected to submit their work on time, late submissions without prior approved excuse will not be accepted.
- Students are required to use report templates if posted by the instructor.
- This course has lectures, assignments, project and Exams.
- The course aims to tackle core concepts, design patterns, and systems that are typically found in real-world video game code. The student will apply such a knowledge to develop a complete game using the Unreal engine.
- The exercises, as well as the assignments in this course, will offer thorough and diverse practices for implementing a complete game by utilizing the core components implemented using the Unreal game engine.
- The student is expected to finish and deliver an answer to the hands-on exercises initiated in each lecture by the next lecture. Participation grades are dependent on such a delivery. Office hours and optional labs are available to provide support as needed.
- Bonus grades are typically attached to some activities and cannot exceed a total of 5%.

EXAMS

As per the below assessment dates table, generally, unless otherwise stated by the instructor, midterm exam/s (if applicable) will be held during the regularly scheduled lecture time.

The topics assessed in the midterm, as well as the final exam, are the core concepts and design patterns utilized in video game architecture. The course project, released exercises, and certain parts of the assignments are tightly coupled to the Unreal Game Engine.

ATTENDANCE AND PUNCTUALITY

Students in this course are required to attend all classes and be on time for all classes. There are no excuses for missing class. All absences are counted as time missed. This applies to all lectures, practical tutorials and/or labs. This also applies to mandatory office hours for probation & Gateway students. Students miss out on opportunities to acquire academic skills such as critical thinking, problem-solving, oral communication, teamwork, and decision-making when they fail to attend class. Being late for class disrupts the flow of the lecture and is disrespectful to the instructors and fellow students. Accordingly, students should note the following:

1. The instructors are not required to give substitute assignments or make up examinations for students who are absent.
2. It is the student's responsibility to make up the class content missed in consultation with his/her instructor(s). This does not mean that the instructor will provide a substitute class for the student.

3. Grades can be affected by missing a class as the instructor may include class participation marks and/or in-class assignments, or pop-up quizzes as part of the assessment plan for the course.
4. If a student is absent for 25% or more of the class time, tutorials, labs, and/or mandatory office hours, the instructor may:
 - a. Advise the student to drop the course if the absences occur before the drop date for course.
 - b. Give the student a grade of F for the course
 - c. Give the student an Incomplete grade for the course if the absences are caused by an approved documented illness or emergency. Students would then be given the opportunity to complete the course work as per the academic regulation.

ACADEMIC INTEGRITY

All students are expected to conduct themselves in all aspects of this course with the highest level of academic integrity. Accordingly, academic dishonesty will not be tolerated. Regulation 20 of the University Calendar defines academic dishonesty. Where there is reasonable evidence to support an allegation of academic dishonesty, Regulation 20 will be strictly enforced. All students are strongly advised to become familiar with this Regulation and UPEI Cairo Academic Integrity policy which can be found at: [!\[\]\(bd1a142de767a21e5362c595f844a4ff_img.jpg\) UPEI Cairo academic integrity policy.pdf](#)

USE OF AI

The use of generative artificial intelligence (AI) and AI-assisted technologies is permitted only **upon receiving a written approval form the instructor**. Students should disclose in their **assignments** the use of AI and AI-assisted technologies in the writing process. Failure to do so will be considered as violation of the academic integrity policy.